

Exploratory Data Analysis (EDA) of Titanic Dataset

Project Overview

This Jupyter Notebook performs comprehensive EDA on the Titanic passenger dataset to uncover patterns and relationships that may explain survival rates.

Tools Used

- Python 3.8+
- Pandas for data manipulation
- Matplotlib and Seaborn for visualization
- Jupyter Notebook for interactive analysis

Dataset Description

The dataset contains information about Titanic passengers with the following columns:

- PassengerId: Unique identifier
- Survived: Survival status (0 = No, 1 = Yes)
- Pclass: Ticket class (1 = 1st, 2 = 2nd, 3 = 3rd)
- Name: Passenger name
- Sex: Passenger gender
- Age: Passenger age
- SibSp: Number of siblings/spouses aboard
- Parch: Number of parents/children aboard
- Ticket: Ticket number
- Fare: Passenger fare
- Cabin: Cabin number
- Embarked: Port of embarkation (C = Cherbourg, Q = Queenstown, S = Southampton)

Key Findings

1. Demographic Insights

- **Age Distribution:** Majority of passengers were between 20-40 years old
- **Gender Ratio:** About 65% male and 35% female passengers
- **Class Distribution:** Over 50% passengers traveled in 3rd class

2. Survival Analysis

- **Overall Survival Rate:** 38.4% of passengers survived

- **Gender Impact:** 74% of female passengers survived vs 19% of males
- **Class Impact:** 63% of 1st class passengers survived vs 24% of 3rd class

3. Family Relationships

- Passengers traveling alone had lower survival rates
- Small families (1-3 members) had better survival chances

How to Use This Notebook

1. Clone the repository
2. Install requirements: `pip install -r requirements.txt`
3. Open and run the Jupyter Notebook
4. View generated visualizations and analysis

Visualizations Included

- Correlation heatmap of numerical features
- Survival rate by passenger class and gender
- Age distribution by survival status
- Fare distribution by passenger class
- Family size impact on survival

Requirements

- `pandas>=1.2.0`
- `matplotlib>=3.3.0`
- `seaborn>=0.11.0`
- `jupyter>=1.0.0`

Future Work

- Feature engineering (title extraction from names)
- Machine learning model for survival prediction
- More detailed family relationship analysis

Author

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